ExpNo12 – Experiment to detect outliers in a given data set.

Program:

```
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

array = np.random.randint(10, 120, 16)
print(array)

print("Mean:", array.mean())
print("25th Percentile:", np.percentile(array, 25))
print("50th Percentile:", np.percentile(array, 50))
print("75th Percentile:", np.percentile(array, 75))
print("100th Percentile:", np.percentile(array, 100))

# Outliers detection
def outDetection(array):
    Q1, Q3 = np.percentile(array, [25, 75])
    IQR = Q3 - Q1
    lr = Q1 - (1.5 * IQR)
    ur = Q3 + (1.5 * IQR)
    return lr, ur

lr, ur = outDetection(array)
```

```
print("Lower Range:", lr)
print("Upper Range:", ur)

sns.histplot(array, kde=True)
plt.show()

new_array = array[(array > lr) & (array < ur)]
print("Array after removing outliers:", new_array)

sns.histplot(new_array, kde=True)
plt.show()

lr1, ur1 = outDetection(new_array)
print("New Lower Range:", lr1)
print("New Upper Range:", ur1)

final_array = new_array[(new_array > lr1) & (new_array < ur1)]
print("Final Array:", final_array)

sns.histplot(final_array, kde=True)
plt.show()
```

Output:

[20 27 12 20 104 86 95 52 11 81 101 46 27 113 85 118]

Mean: 62.375

25th Percentile: 25.25

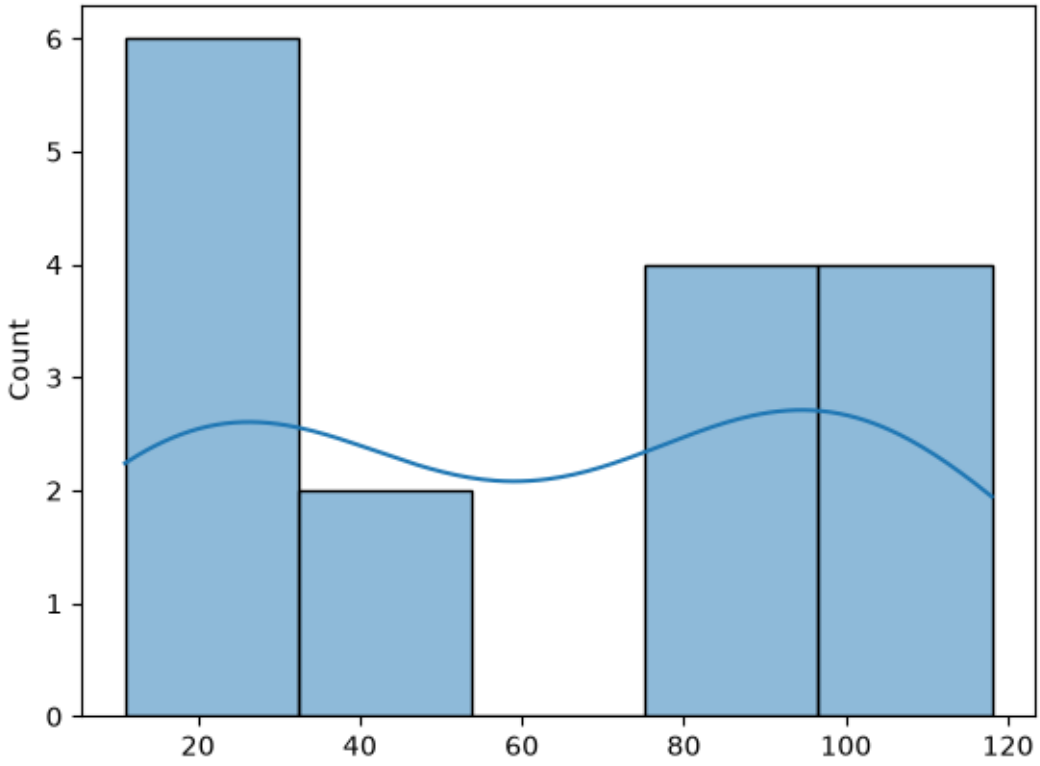
50th Percentile: 66.5

75th Percentile: 96.5

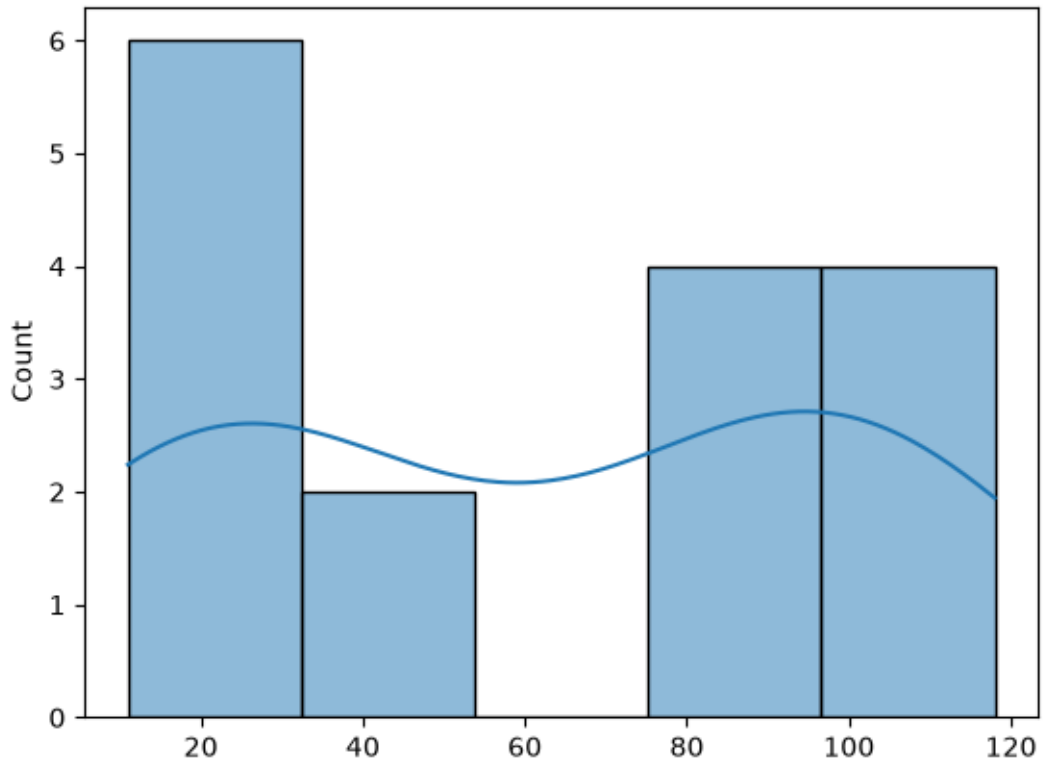
100th Percentile: 118.0

Lower Range: -81.625

Upper Range: 203.375



Array after removing outliers: [20 27 12 20 104 86 95 52 11 81 101 46 27 113 85 118]



New Lower Range: -81.625

New Upper Range: 203.375

Final Array: [20 27 12 20 104 86 95 52 11 81 101 46 27 113 85 118]

